

Customer Churn Analysis

Telecom Subscription Company · End-to-End Analysis using Python & Power BI

Dataset: 7,043 customer records **Features:** 21 columns **Target:** Churn (Yes / No) **Analysis Period:** Tenure 1–72 months

1. Tools & Technologies

Python 3.10+

Pandas

NumPy

Matplotlib

Seaborn

Jupyter Notebook

Power BI Desktop

Power Query

DAX

2. Data Cleaning Summary

ISSUES FOUND & RESOLVED

Duplicate rows	5 removed
Missing — Monthly_Charges	10 → median imputed
Missing — Total_Charges	10 → tenure × charge
Data type corrections	16 columns cast
Tenure groups created	6 bands (0–72 months)

FINAL CLEANED DATASET

Total rows	7,043
Total columns	22 (incl. Tenure_Group)
Missing values	0
Duplicate rows	0
Exported as	Customer_Churn_Cleaned.csv

3. Key Performance Indicators

7,043

TOTAL CUSTOMERS

2,415

CHURNED CUSTOMERS

4,628

RETAINED CUSTOMERS

34.29%

OVERALL CHURN RATE

\$60.26

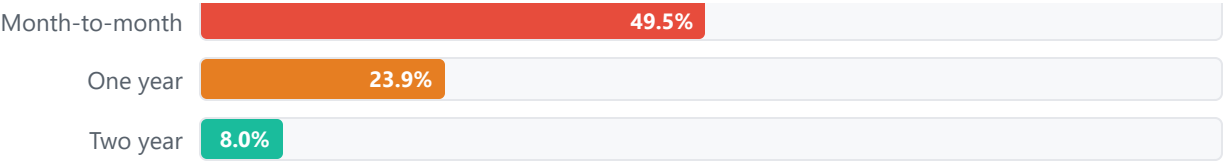
AVG MONTHLY CHARGES

36.36

AVG TENURE (MONTHS)

4. Churn Rate by Customer Segment

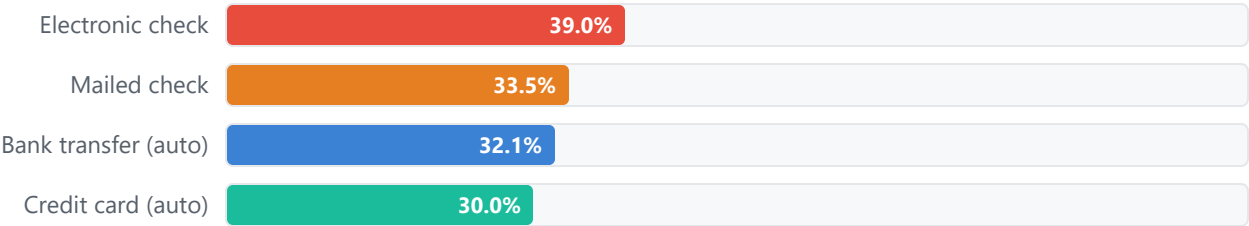
BY CONTRACT TYPE



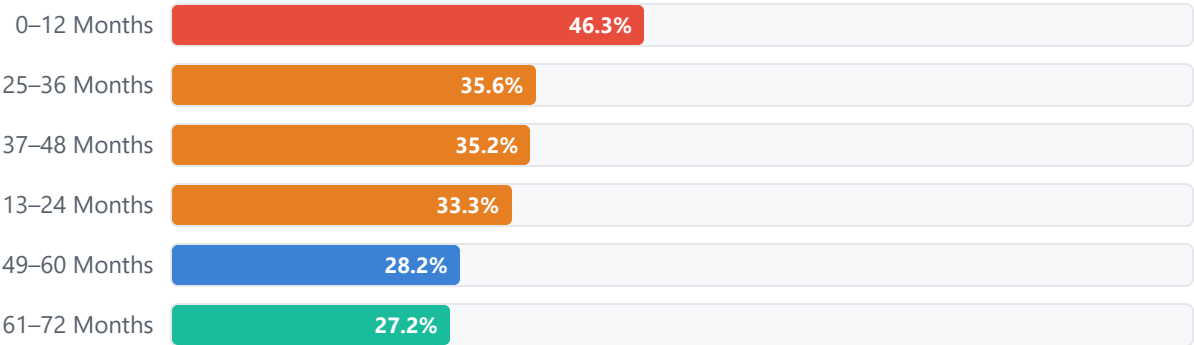
BY INTERNET SERVICE



BY PAYMENT METHOD



BY TENURE GROUP



BY GENDER & SENIOR CITIZEN STATUS

GENDER		SENIOR CITIZEN STATUS	
Male	34.54%	Senior (1)	38.58%
Female	34.04%	Non-Senior (0)	33.51%
Gender has minimal impact (<1% difference)		Seniors churn ~5% more than non-seniors	

5. Churned vs Retained — Comparative Analysis

Metric	Churned	Retained	Difference
Count	2,415 34.3%	4,628 65.7%	—
Avg Monthly Charges	\$63.34	\$58.65	+\$4.69 higher
Avg Tenure (Months)	33.1	38.0	−4.9 months shorter
Dominant Contract	Month-to-month	Two year	Strong predictor
Dominant Internet	Fiber optic	DSL / No internet	Fiber 41.7% vs DSL 27.8%
Dominant Payment	Electronic check	Auto payment methods	Manual pay = higher risk

6. Python Analysis — Coverage Summary

- ✓ Part 1: Data Loading (Steps 1–6)

✓ Part 3: Analysis — 22 metrics (Steps 17–38)

✓ Churn Distribution Pie Chart

✓ Churn by Contract, Gender, Internet, Payment

✓ Tenure vs Monthly Charges Scatter Plot

✓ Churn Rate by Tenure Group Bar Chart

✓ Feature Correlation Heatmap (17 features)
- ✓ Part 2: Data Cleaning (Steps 7–16)

✓ Part 4: 14 Visualisations created

✓ Contract Distribution Bar Chart

✓ Tenure & Monthly Charges Histograms

✓ Monthly & Total Charges Box Plots

✓ Service Usage Count Plots (8 services)

✓ Part 5: Business Insights (10 insights)

7. Power BI Dashboard — Coverage

Component	Items	Status
KPI Cards	Total Customers, Churned, Retained, Churn Rate %, Avg Monthly Charges, Avg Tenure	6 ✓
DAX Measures	COUNTROWS, CALCULATE, DIVIDE, AVERAGE — 6 measures	6 ✓
Visuals	Donut, Bar (×3), Column, Box/Column, Scatter, Table	10 ✓
Slicers	Gender, Contract, Internet Service, Payment Method, Senior Citizen, Churn	6 ✓

1	The overall churn rate is	34.29%	— significantly above the 15–20% telecom industry benchmark, signalling an urgent retention problem.
2	Month-to-month contract customers churn at 49.5%		— nearly 3× the rate of one-year (23.9%) and 6× the rate of two-year (8.0%) contract holders.
3	Fiber optic internet users churn at 41.7%		, the highest of any internet category — suggesting a mismatch between price expectations and perceived service quality.
4	Electronic check payers churn at 39.0%		, the highest payment method, compared to ~30–32% for automatic payment customers.
5	New customers (0–12 months) are most vulnerable		— 46.3% churn rate in the first year. The first 12 months represent the highest-risk window.
6	Churned customers pay on average	\$4.69 more per month	(\$63.34 vs \$58.65) — price sensitivity is a key driver of churn.
7	Senior citizens churn 5% more		(38.6% vs 33.5%) — dedicated support packages and personalised retention outreach are recommended.
8	Customers without add-on services		(Online Security, Tech Support) show significantly higher churn — bundling acts as a natural retention mechanism.
9	Long-tenure customers (61–72 months) churn at only	27.2%	— loyalty programmes and early lock-in incentives could move more customers into this stable zone.
10	The highest-risk segment		is: Month-to-month + Fiber optic + Senior Citizen — primary target for proactive retention campaigns.

FOCUS AREA	RECOMMENDATION	PRIORITY
Contract conversion	Offer month-to-month customers discounts to upgrade to 1–2 year contracts	High
Fiber optic value	Review Fiber pricing or enhance service quality to justify premium billing	High

FOCUS AREA	RECOMMENDATION	PRIORITY
Payment automation	Incentivise customers to switch from manual to auto-pay to reduce friction	High
Early retention (0–12 months)	Deploy onboarding programmes and first-year discounts to reduce early churn	Medium
Senior citizen segment	Create tailored plans with simplified billing and dedicated support lines	Medium
Service bundling	Promote Online Security and Tech Support bundles — add-on users retain better	Standard

10. Project Deliverables

- ✓

 Customer_Churn_Raw.csv
- ✓

 Customer_Churn_Analysis.ipynb
- ✓

 14 Visualisation PNGs exported
- ✓

 Project Report PDF
- ✓

 Customer_Churn_Cleaned.csv
- ✓

 Customer_Churn_Dashboard.pbix
- ✓

 README.md
- ✓

 6 DAX Measures documented